

Introduction to Random Tilings

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I. INTRODUCTION

- The goal of this talk is to give an introduction to the study of random tilings. The general setup for tiling problems consists of three main ingredients: a lattice, a domain within this lattice, and a collection of elementary domains called “tiles”. A “tiling” of the domain $\mathcal{R}$ is then any partition of $\mathcal{R}$ via these tiles.
- In this talk, we shall focus on the case in which the lattice is a triangular grid of unital mesh (more specifically, the lattice generated by the vectors $(0, 1)$ and $(-\frac{\sqrt{3}}{2}, -\frac{1}{2})$), the domain $\mathcal{R}$ is a polygonal domain in this grid, and in which the tiles are any one of three kinds of rhombi (which we refer to as “lozenges”).

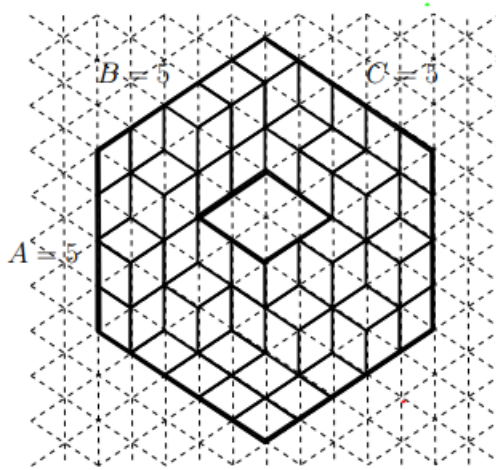


Figure 1: a lozenge tiling of a $5 \times 5 \times 5$ hexagon with a hole

II. MOTIVATION

- There are several reasons why one might wish to study tilings. As a pure mathematician, my personal favourite is the interesting mathematics itself. For instance, consider the following beautiful figure, which depicts a “typical” (i.e. uniformly random) lozenge tiling of a large hexagon. Observe that near the boundary of the domain, one sees regions consisting of only a single type of lozenge. Such regions are called “frozen regions” (for obvious reasons) and their boundaries “arctic curves” (for similarly obvious reasons). It hopefully shouldn’t take too much convincing that the study of frozen regions and their arctic circles (and the asymptotic phenomena thereof) is incredibly deep; in fact, it is the driving force behind much of the study of random tilings. There are, however, a multitude of non-pure-mathematical motivations for the study of tilings. Most of these come from statistical mechanics. For instance, without going into too much detail, one can relate Lozenge tilings to the Ising and 6-vertex models.

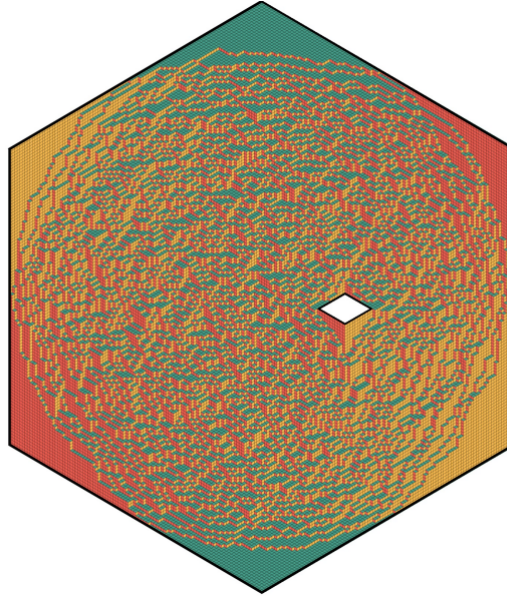


Figure 2: uniformly random lozenge tiling of a large hexagon with a hole

III. MATHEMATICAL QUESTIONS

- What kind of questions shall drive our study of tilings? The first and most obvious is the question of existence, that is, given a domain $\mathcal{R}$ drawn on the triangular grid, does there exist a tiling thereof? Moreover, can one determine this in a simple and efficient (say, in polynomial time) manner? Answering this question shall be the main focus of this talk. One easily sees (give an example) that some domains are tileable in more than one way, while others (give an example) are not tileable at all.
- The latter begs the question: given a domain $\mathcal{R}$, can we count how many tilings $\mathcal{R}$ has? More specifically, can we find a “closed form” (i.e. non-recursive) explicit formula for this number? The answer to this question is (mostly) yes, though whether many of the formulas are truly “explicit” depends on one’s personal taste (for instance, they are often given by determinants of certain matrices as opposed to formulas that can be evaluated “by hand”).
- Up to rescaling, lozenges can be thought of as the projections of the sides of a unit cube onto the $x + y + z = 0$ plane. Under this perspective, a tiling can be viewed as the projection of a 2-dimensional stepped surface onto the $x + y + z = 0$ plane, and the function h whose graph is this surface is called the *height function* associated to the tiling (we’ll talk about these in more detail later). The upshot is that if we take a uniformly random tiling of a given domain, then we obtain a random corresponding height function h , which as per our previous remarks encodes a random stepped surface. A natural question to ask is whether a law of large numbers holds for this random function h . As it turns out, for a domain of linear size L , the rescaled height function $L^{-1}h(Lx, Ly)$ converges to a deterministic limit $\hat{h}(x, y)$ as $L \rightarrow \infty$. One of the main pursuits within the study of random tilings is how to compute the so-called “limit shape” $\hat{h}(x, y)$.
- Another natural question to ask is whether a central limit theorem holds for the fluctuations of the random function h . Again, it turns out that for a domain of linear size L , the limit

$$\lim_{L \rightarrow \infty} (h(Lx, Ly) - \mathbb{E}[h(Lx, Ly)])$$

exists (in fact, it is given by the Gaussian free field, which is essentially a higher dimensional analogue of Brownian motion). One might object that the scaling above is unusual: perhaps if we divided out by $\sqrt{L}$ it would look more like the classical central limit theorem. It turns out that there is some level of “rigidity” in tilings and that the fluctuations are much smaller, so this re-normalization is actually not necessary.

- Finally, one might be interested in how the arctic curves (i.e. the borders of the frozen region) fluctuate as the size of the domain increases. What is the correct scaling which characterizes the growth of these fluctuations? It turns out (and this is the connection to KPZ) to be $L^{1/3}$, and that the asymptotic law of the rescaled fluctuations is the Tracy-Widom distribution.

III. THURSTON'S THEOREM ON TILEABILITY

- Here, we begin our study of the first question: given a domain $\mathcal{R}$ drawn on the triangular grid, does there exist a tiling thereof, and can we determine this in a computationally efficient manner?
- First, we observe that without any loss of generality we may assume that the domain $\mathcal{R}$ is connected, since (clearly) $\mathcal{R}$ is tileable if and only if each of its connected components are. To kick off our analysis, we shall also make the stronger assumption that $\mathcal{R}$ is in fact simply connected, and then later remove this hypothesis.
- Next, because it shall be relevant to the main result of this talk, we define an asymmetric non-negative distance function $d(u, v)$ on the vertices of $\mathcal{R}$ by declaring $d(u, v)$ to be the minimal number of edges in a positively oriented path from u to v which stays within or on the boundary of $\mathcal{R}$ (here, by “positively oriented” we mean “consists entirely of positive edges”). Note that this definition makes sense (i) because of our connectedness assumption and (ii) because each one of the negative directions $-a$, $-b$, and $-c$ can be written as a sum of positive directions. Note, moreover, that because of orientation restriction, the function d is asymmetric. For instance, if $\mathcal{R}$ is a single triangle and u and v distinct vertices thereof, then $\{d(u, v), d(v, u)\} = \{1, 2\}$.
- There is now only one more thing we need to do before stating the main result, which is to give a more detailed discussion of the notion of a height function introduced earlier. Observe that within the triangular grid, there are six directions (draw a picture). The unit vectors in these directions are

$$a = (0, 1), \quad b = \left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right), \quad c = \left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right), \quad -a, \quad -b, \quad -c.$$

By convention, we call the directions a , b , and c *positive* and their negations *negative*. Given a tiling of $\mathcal{R}$, the associated *height function* h , which acts on the vertices $v \in \mathcal{R}$, is defined via the following local rule: for any vertex u such that v is positively adjacent to u (by this, we mean that u should be connected to v via an edge, and that if we walk along this edge from u towards v , we walk in one of the three positive directions), we have that

$$h(v) - h(u) = \begin{cases} 1 & \text{if we follow an edge of a lozenge in moving from } u \text{ to } v \\ -2 & \text{if we cross a lozenge diagonally in moving from } u \text{ to } v \end{cases}.$$

- Of course, one must check that this construction actually yields a well-defined function h . This, however, is an easy task. Namely, it is straightforward to check that h is “defined consistently” for a single lozenge, and that the definition “extends consistently” across unions. As tilings are just unions of lozenges, this tells us that a well-defined height function exists for any tiling. Notice that because the local rule by which h is defined is a difference $h(v) - h(u)$, the function h is only determined up to a constant shift. This allows us to assume that $h(v_0) = 0$ for our favourite vertex v_0 (and in fact we have to make a choice of shift in order for h to be defined at all).
- Why are height functions named as such? We previously alluded to the idea that tilings can be thought of as the projections of stepped surfaces, and that these stepped surfaces are the graphs of the corresponding height functions. It is natural to then speculate that for a fixed tiling of $\mathcal{R}$, the value of $h(v)$ has some connection to the distance between v and the corresponding stepped surface. It turns out that this is indeed the case. Without going into too much detail, one can check that $h(v)$ equals the signed and rescaled distance from the v to its corresponding point on the stepped surface.
- The utility of the simply connected hypothesis is that even if $\mathcal{R}$ is not tileable, our local rules specify h uniquely (up to a global shift, of course) on the boundary $\partial\mathcal{R}$. This is because we can first pick a vertex on $\partial\mathcal{R}$ to have height zero (this specifies our global shift), and then note that lozenges

cannot cross $\partial\mathcal{R}$ in any tiling thereof, so h must of increments of $+1$ in all positive directions along $\partial\mathcal{R}$ and increments of -1 in all negative directions along $\partial\mathcal{R}$. This immediately gives us a necessary condition for the existence of a tiling.

- It is not too difficult to see, however, that this condition is not sufficient for the existence of a tiling (give an example). Roughly, the problem is that vertices can be “too close”, forcing a particular type of lozenge and causing obstruction.

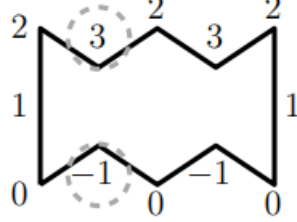


Figure 3: A non-tilable domain for which h is defined consistently on the boundary.

- By imposing a condition preventing the above problem, we get the main result of this talk:

Theorem (Thurston, 1990). *Let $\mathcal{R}$ be a simply connected domain in the triangular lattice. Then $\mathcal{R}$ is tileable if and only if both of the following hold:*

1. *One can define h on $\partial\mathcal{R}$ so that $h(v) - h(u) = 1$ whenever v is positively adjacent to u , and*
2. *This h has the property that for any $u, v \in \partial\mathcal{R}$,*

$$h(v) - h(u) \leq d(u, v).$$

IV. PROOF OF THURSTON’S THEOREM

- Suppose first that $\mathcal{R}$ is tileable. We must show that the two conditions in the theorem are satisfied. From our discussion leading up to the theorem, we already know that the first condition holds; therefore, we just need to check the second. Thankfully, this isn’t too hard. Given distinct vertices $u, v \in \partial\mathcal{R}$, choose any positively oriented path from u to v , and label the vertices thereof $u = w_0, w_1, \dots, w_{n-1}, w_n = v$. Then, by definition, the increments of h along this path are always either $+1$ or -2 . On the other hand, those of d are (clearly) always $+1$. Therefore,

$$h(v) - h(u) = \sum_{i=1}^n [h(w_i) - h(w_{i-1})] \leq \sum_{i=1}^n [d(u, w_i) - d(u, w_{i-1})] = d(u, v)$$

as desired.

- Conversely, suppose now that the two conditions of the theorem are satisfied. We wish to show that $\mathcal{R}$ is tileable; we do so by using the conditions to construct one possible identity of h and then reverse engineering the corresponding tiling. The key observation is the following: the second condition in the theorem can be re-written as

$$\forall v \in \partial\mathcal{R}, \quad h(v) \leq \min_{u \in \partial\mathcal{R}} [d(u, v) + h(u)].$$

One can then naively guess a possible identity of h by declaring $h(v) = \min_{u \in \partial\mathcal{R}} [d(u, v) + h(u)]$ for all $v \in \mathcal{R}$. In light of the above, we call this the “maximal height function” extending $h|_{\partial\mathcal{R}}$ (which is given to us up to a global shift). Note that it is straightforward to see that this definition of h has the required unit increments along the positive edges of the boundary.

- To complete the proof, it is now enough to check that this h changes by either $+1$ or -2 along each positive edge. Roughly, this is because this would tell us that h is indeed a height function, and so must correspond to some tiling. More precisely, we can explicitly reverse engineer a tiling by enforcing the local rules defined earlier. In particular, as per these rules, we assume each -2 edge to be the diagonal of a lozenge; this then uniquely determines the lozenge to which each subtriangle of $\mathcal{R}$ belongs, thereby yielding a tiling.

- Let $v \rightarrow w$ be a positively oriented edge in $\mathcal{R} \setminus \partial\mathcal{R}$ (we already know that h changes by either $+1$ or -2 along positive edges of the boundary; in fact, it's always the former). We begin by establishing two useful estimates that will simplify things quite a bit. The first is that for any $u \in \partial\mathcal{R}$, we have that $d(u, w) \leq d(u, v) + 1$ by augmenting a minimal (u, v) -path by a single edge. Therefore,

$$h(w) = \min_{u \in \partial\mathcal{R}} [d(u, w) + h(u)] \leq \min_{u \in \partial\mathcal{R}} [d(u, v) + h(u)] + 1 = h(v) + 1.$$

Similarly, by augmenting a minimal (u, w) -path by two positive edges we get that $d(u, v) \leq d(u, w) + 2$ and hence

$$h(w) \geq h(v) - 2.$$

It follows that $h(w) - h(v) \in \{-2, -1, 0, 1\}$, so we just need to rule out the values 0 and -1 . We do this via two observations (which we state first and then prove).

- The first is that for any two positively oriented paths between the same two vertices, their number of edges differ by a multiple of three. The second is that the value of $d(u, v) + h(u)$ changes by multiples of three as u varies over $\partial\mathcal{R}$. To see that the conjunction of these observations yields the result, note that by augmenting a minimal (u, v) -path by a single edge and applying the first observation, we get (for some $u \in \partial\mathcal{R}$) that

$$h(w) = d(u, w) + h(u) \equiv d(u, v) + 1 + h(u) \pmod{3}.$$

By the second observation, the right-hand side is furthermore congruent to $h(v) + 1 \pmod{3}$, so we have that $h(w) - h(v) \equiv 1 \pmod{3}$ and hence $h(w) - h(v) \notin \{-1, 0\}$.

- It remains only to prove these observations. In illustration of the first, we let n_a^1, n_b^1, n_c^1 and n_a^2, n_b^2, n_c^2 denote the numbers of edges of the three types in each of the two paths and project onto the a and b directions to obtain

$$2n_a^1 - n_b^1 - n_c^1 = 2n_a^2 - n_b^2 - n_c^2 \quad \text{and} \quad -n_a^1 + 2n_b^1 - n_c^1 = -n_a^2 + 2n_b^2 - n_c^2.$$

Adding the above identities then yields $(n_a^1 + n_b^1 + n_c^1) - (n_a^2 + n_b^2 + n_c^2) = 3(n_c^1 - n_c^2)$. In illustration of the second observation, we note that by connectivity of $\partial\mathcal{R}$ (recall that $\mathcal{R}$ is simply connected) it is enough to check that $d(u, v) + h(u)$ changes by a multiple of three as u moves along a single edge in $\partial\mathcal{R}$. This follows from the first observation together with what we know about the increments of h along $\partial\mathcal{R}$. This completes the proof of Thurston's theorem.

V. GENERALIZING THURSTON'S THEOREM

- We would like to generalize this result to non-simply-connected domains. There are, however, a few issues with this. The first is that it is not clear how h should be defined on $\partial\mathcal{R}$ if one is not already given a tiling. This is because $\partial\mathcal{R}$ a priori has several connected components, and although h is uniquely determined up to a constant shift on each one, these constants may differ across connected components. The second is that one of the key steps in the proof was that we could determine the variation of $d(u, v) + h(u)$ just by moving along $\partial\mathcal{R}$ edge by edge. In multiply connected domains, we obviously cannot do this.
- It turns out that these issues can be remedied by requiring that $d(u, v) - h(v) + h(u) \equiv 0 \pmod{3}$ for all $u, v \in \mathcal{R}$, and by simply leaving the ambiguity in h untouched:

Corollary. *Let $\mathcal{R}$ be a connected (but not necessarily simply-connected) domain in the triangular lattice. Define h along $\partial\mathcal{R}$ uniquely up to constants $c_1, \dots, c_k$ corresponding to shifts along the k connected components of $\partial\mathcal{R}$. Then $\mathcal{R}$ is tileable if and only if one can choose the constants $c_1, \dots, c_k$ so that for all $u, v \in \partial\mathcal{R}$ we have that*

$$d(u, v) - h(v) + h(u) \geq 0 \quad \text{and} \quad d(u, v) - h(v) + h(u) \equiv 0 \pmod{3}.$$

- The proof the above just boils down to the fact that the conditions ensure that our original proof carries over. To be specific, the first condition ensures that the proof of the forward direction carries over and the second condition ensures that the proof of the backwards direction carries over.